

Lesson 02 Demo 02

Implementing Encryption for CodePipeline Workflow

Objective: To demonstrate the process of implementing encryption in an AWS CodePipeline workflow, ensuring secure handling of artifacts, source code, and environment variables using AWS Key Management Service (KMS)

Tools required: AWS Lab, GitHub

Prerequisites: None

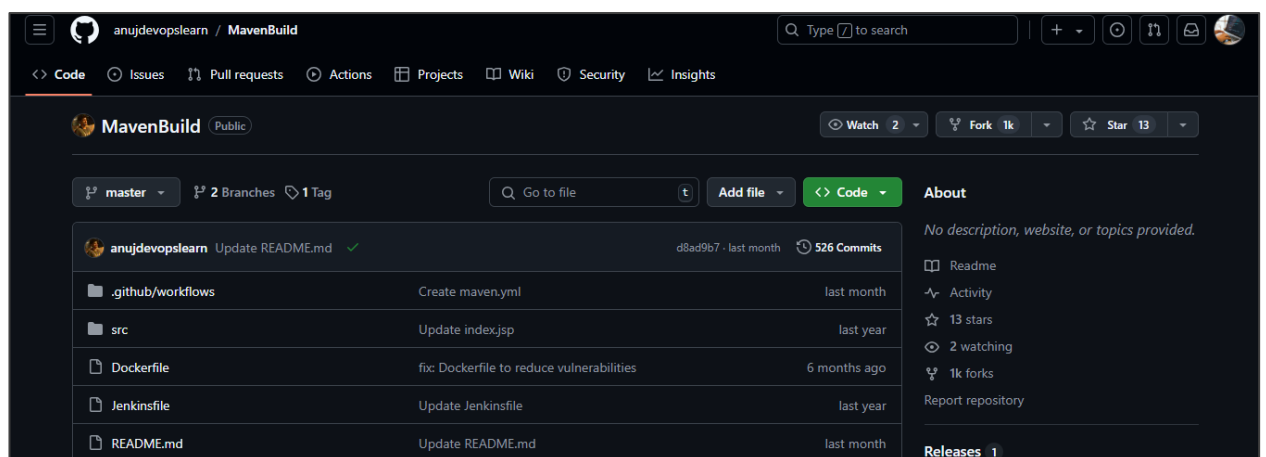
Steps to be followed:

1. Create an S3 bucket with a KMS key
2. Create a Continuous Integration pipeline
3. Edit the pipeline and add the Artifacts section
4. Test the updated pipeline

Step 1: Create an S3 bucket with a KMS key

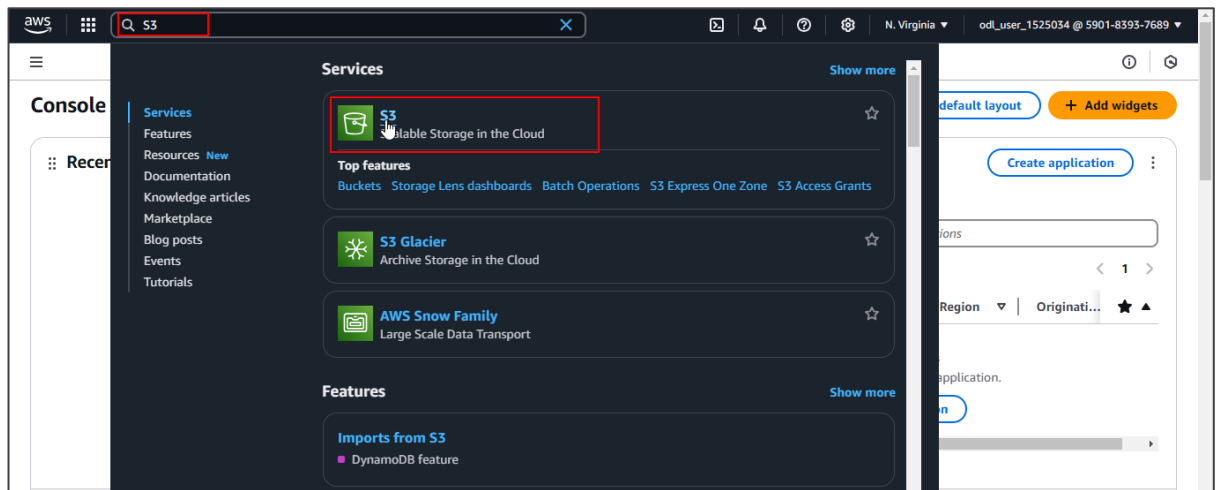
1.1 Log in to your GitHub account and import the below repository:

<https://github.com/anujdevopslearn/MavenBuild>

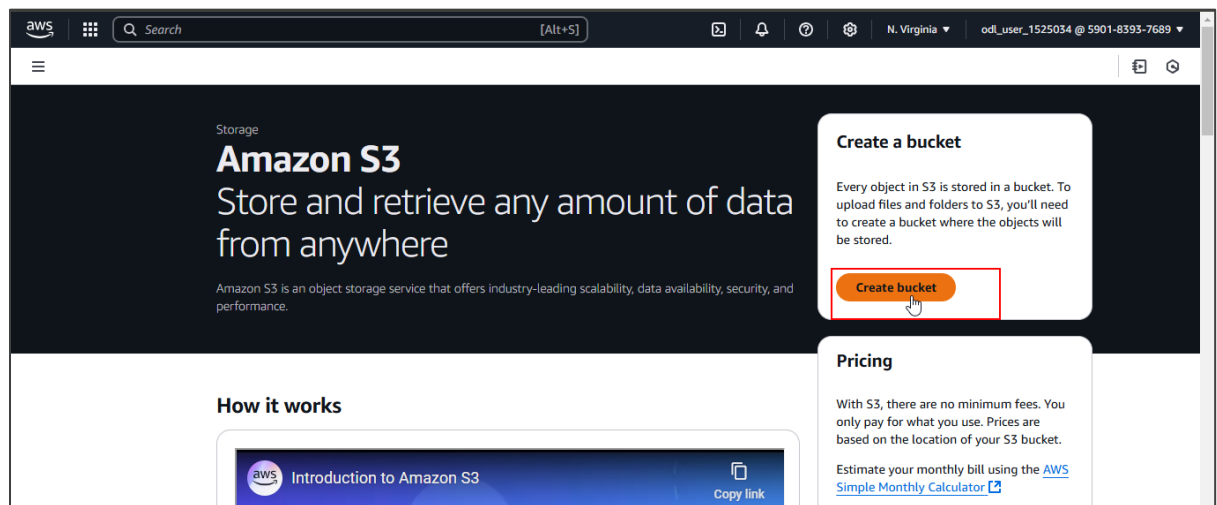


Once the repository is imported, you must create an S3 bucket to store encrypted artifacts.

1.2 Open the AWS dashboard, search for and select the S3 service



1.3 Click on Create bucket



1.4 Select the **Bucket type** as **General purpose** and give the **Bucket name** as **codepipelineartifactbucket123**

The screenshot shows the AWS console 'Create bucket' page. The 'General configuration' section is visible. The 'AWS Region' is 'US East (N. Virginia) us-east-1'. The 'Bucket type' is set to 'General purpose' (selected with a radio button). The 'Bucket name' is 'codepipelineartifactbucket123' (highlighted with a red box). Below the bucket name, there is a note: 'Bucket name must be unique within the global namespace and follow the bucket naming rules. [See rules for bucket naming](#)'. There is also a 'Copy settings from existing bucket - optional' section with a 'Choose bucket' button. The format 's3://bucket/prefix' is shown at the bottom.

Note: Make sure you provide a unique bucket name which is not used earlier

1.5 Scroll down to the **Default encryption** section, select the **Encryption type** as **Server-side encryption with AWS Key Management Service keys (SSE-KMS)**, and specify the **AWS KMS key** by selecting the **Enter AWS KMS key ARN**

The screenshot shows the AWS console 'Create bucket' page, scrolled down to the 'Default encryption' section. The 'Encryption type' is set to 'Server-side encryption with AWS Key Management Service keys (SSE-KMS)' (selected with a radio button and highlighted with a red box). The 'AWS KMS key' is set to 'Enter AWS KMS key ARN' (selected with a radio button and highlighted with a red box). Below this, the 'AWS KMS key ARN' field is shown with a placeholder 'arn:aws:kms:<region>:<account-ID>:key/<key-id>' and a 'Create a KMS key' button. The 'Bucket Key' section at the bottom shows 'Enable' selected.

1.6 Click on **Create a KMS key**

The screenshot shows the AWS console 'Create bucket' page. Under the 'Default encryption' section, the 'Encryption type' is set to 'Server-side encryption with AWS Key Management Service keys (SSE-KMS)'. Under the 'AWS KMS key' section, the option 'Enter AWS KMS key ARN' is selected. A text input field for the ARN is shown with a placeholder format: `arn:aws:kms:<region>:<account-ID>:key/<key-id>`. A red box highlights the 'Create a KMS key' button next to the input field. Below the input field, the format for the key ID is explained: `arn:aws:kms:<region>:<account-ID>:key/<key-id>` (using key id) and `arn:aws:kms:<region>:<account-ID>:alias/<alias-name>` (using alias). The 'Bucket Key' section at the bottom has 'Enable' selected.

1.7 Keep the **Key type** as **Symmetric** and the **Key usage** as **Encrypt and decrypt** under the **Configure key** section. Then, click on **Next**.

The screenshot shows the 'Configure key' page in the AWS console. On the left, a navigation pane lists steps: 'Configure key' (selected), 'Add labels', 'Define key administrative permissions', 'Define key usage permissions', and 'Review'. The main content area has two sections: 'Key type' and 'Key usage'. In the 'Key type' section, 'Symmetric' is selected with a red box around it. In the 'Key usage' section, 'Encrypt and decrypt' is selected with a red box around it. At the bottom right, there are 'Cancel' and 'Next' buttons, with the 'Next' button highlighted by a red box.

1.8 Enter the **Alias** as **artifactskey** under the **Add labels** section

The screenshot shows the AWS KMS 'Create key' wizard at Step 2, 'Add labels'. The left sidebar shows the progress: Step 1 (Configure key), Step 2 (Add labels), Step 3 (Define key administrative permissions), Step 4 (Define key usage permissions), Step 5, and Review. The main content area has a title 'Add labels' and three sections: 'Alias', 'Description - optional', and 'Tags - optional'. The 'Alias' section has a text input field containing 'artifactskey'. The 'Description' section has a text area with the placeholder 'Description of the key'. The 'Tags' section is currently empty.

Alias
You can change the alias at any time. [Learn more](#)

Alias
artifactskey

Description - optional
You can change the description at any time.

Description
Description of the key

Tags - optional

1.9 Scroll down and click on **Next**

The screenshot shows the AWS KMS 'Create key' wizard at Step 5, 'Review'. The left sidebar shows the progress: Step 1 (Configure key), Step 2 (Add labels), Step 3 (Define key administrative permissions), Step 4 (Define key usage permissions), Step 5 (Review), and Review. The main content area has a title 'Review' and two sections: 'Description - optional' and 'Tags - optional'. The 'Description' section has a text area with the placeholder 'Description of the key'. The 'Tags' section has a text area with the placeholder 'This key has no tags.' and an 'Add tag' button. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Next'. The 'Next' button is highlighted with a red box and a mouse cursor.

Description - optional
You can change the description at any time.

Description
Description of the key

Tags - optional
You can use tags to categorize and identify your KMS keys and help you track your AWS costs. When you add tags to AWS resources, AWS generates a cost allocation report for each tag. [Learn more](#)
This key has no tags.

[Add tag](#)
You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Next](#)

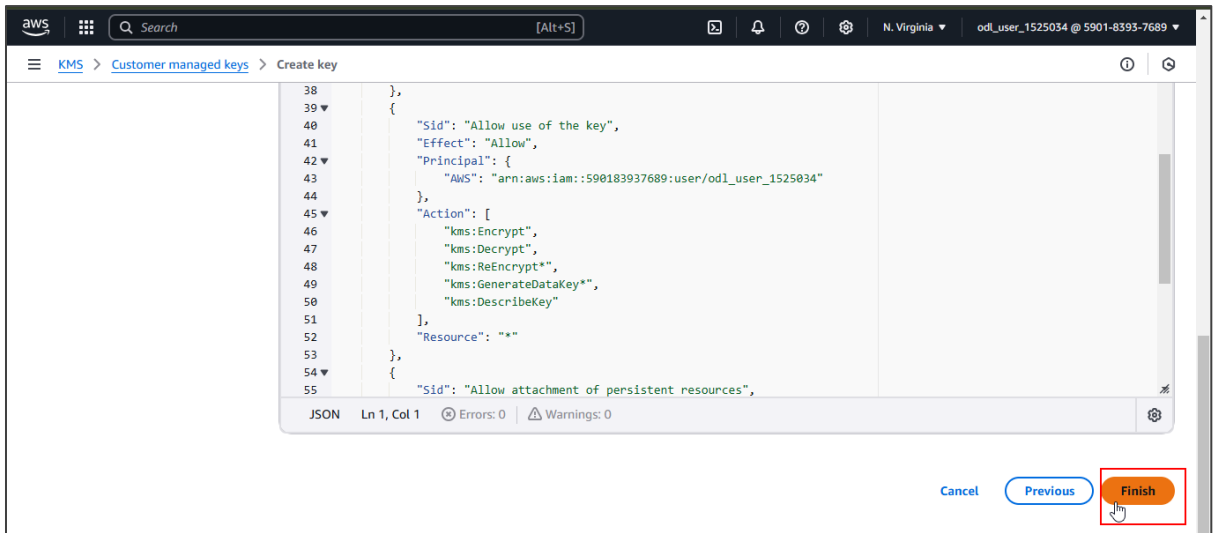
- 1.10 Search for the **odl_user**, select the checkbox for the respective **odl_user**, and under the **Define key administrative permission section**, select the checkbox for **Allow key administrators to delete this key**. Then, click **Next**.

The screenshot shows the AWS Management Console interface for the 'Create key' process, specifically the 'Define key administrative permissions' step. The left-hand navigation pane shows five steps: 'Step 1: Configure key', 'Step 2: Add labels', 'Step 3: Define key administrative permissions' (which is the active step), 'Step 4: Define key usage permissions', and 'Step 5: Review'. The main content area is titled 'Define key administrative permissions'. It contains two sections: 'Key administrators (1/17)' and 'Key deletion'. In the 'Key administrators' section, a search bar contains 'odl_user' and shows '1 matches'. Below the search bar is a table with columns 'Name', 'Path', and 'Type'. The table has one row with 'odl_user_1525034' in the 'Name' column, '/' in the 'Path' column, and 'User' in the 'Type' column. A checkbox is checked next to the 'Name' column header, and another checkbox is checked next to the 'odl_user_1525034' row. In the 'Key deletion' section, a checkbox labeled 'Allow key administrators to delete this key.' is checked. At the bottom right of the console, there are three buttons: 'Cancel', 'Previous', and 'Next'. The 'Next' button is highlighted with a red box and a mouse cursor.

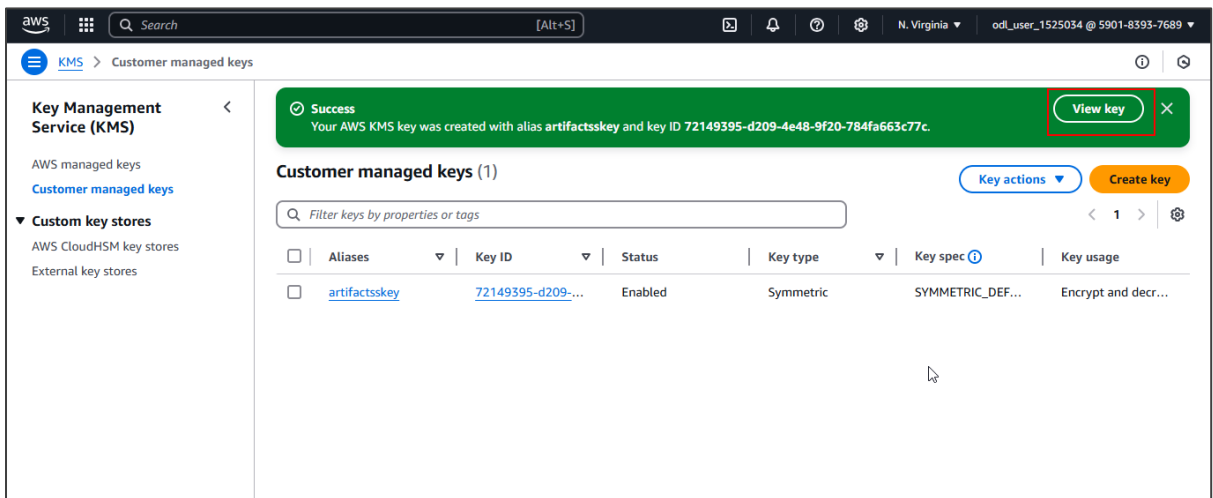
- 1.11 Search for the **odl_user**, select the checkbox for **odl_user_1525034** under **Key Users**, and click **Next**

The screenshot shows the AWS Management Console interface for the 'Create key' process, specifically the 'Define key usage permissions' step. The left-hand navigation pane shows five steps: 'Step 2: Add labels', 'Step 3: Define key administrative permissions', 'Step 4: Define key usage permissions' (which is the active step), 'Step 5: Review', and 'Step 6: Review'. The main content area is titled 'Define key usage permissions'. It contains two sections: 'Key users (1/17)' and 'Other AWS accounts'. In the 'Key users' section, a search bar contains 'odl_user' and shows '1 matches'. Below the search bar is a table with columns 'Name', 'Path', and 'Type'. The table has one row with 'odl_user_1525034' in the 'Name' column, '/' in the 'Path' column, and 'User' in the 'Type' column. A checkbox is checked next to the 'Name' column header, and another checkbox is checked next to the 'odl_user_1525034' row. In the 'Other AWS accounts' section, there is a button labeled 'Add another AWS account'. At the bottom right of the console, there are three buttons: 'Cancel', 'Previous', and 'Next'. The 'Next' button is highlighted with a red box and a mouse cursor.

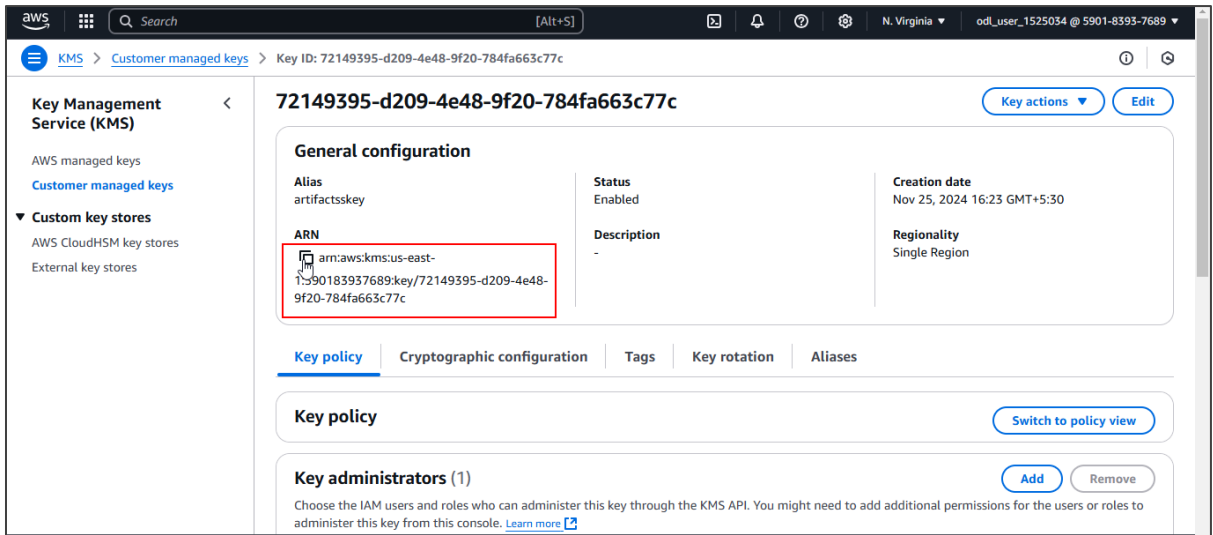
1.12 Scroll down and click **Finish**



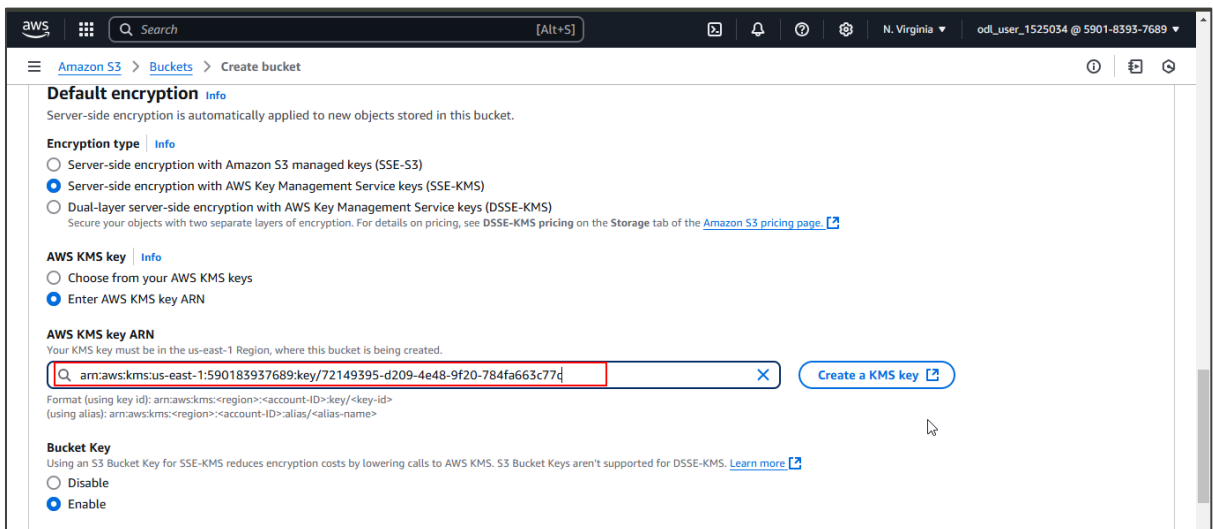
1.13 Click on **View key** after successfully creating the AWS KMS key



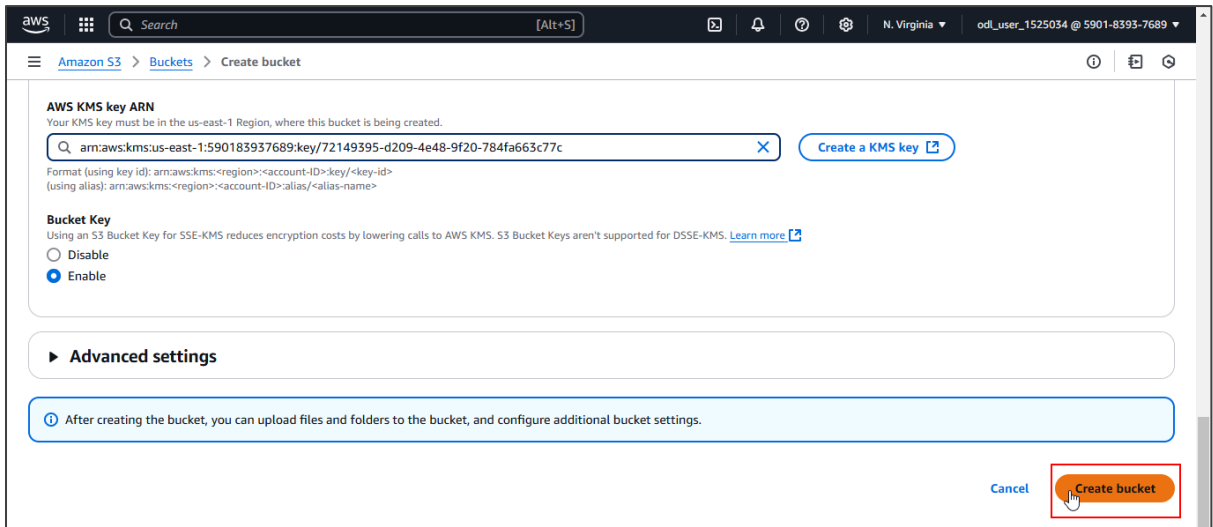
1.14 Copy the ARN from the General configuration section



1.15 Navigate back to the Buckets page and enter the copied ARN in the AWS KMS key ARN section



1.16 Scroll down and click on **Create bucket**



aws Search [Alt+S] N. Virginia odl_user_1525034 @ 5901-8393-7689

Amazon S3 > Buckets > Create bucket

AWS KMS key ARN
Your KMS key must be in the us-east-1 Region, where this bucket is being created.

arn:aws:kms:us-east-1:590183937689:key/72149395-d209-4e48-9f20-784fa663c77c

Create a KMS key

Format (using key id): arn:aws:kms:<region>:<account-ID>:key/<key-id>
(using alias): arn:aws:kms:<region>:<account-ID>:alias/<alias-name>

Bucket Key
Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

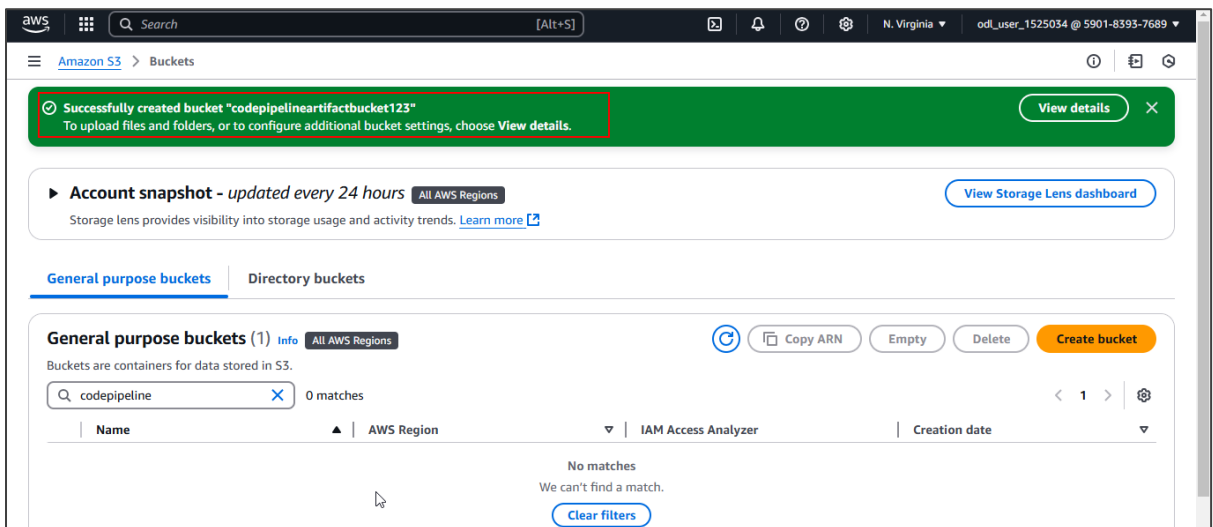
☐ Disable
☒ Enable

Advanced settings

After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.

Cancel Create bucket

The bucket is created successfully.



aws Search [Alt+S] N. Virginia odl_user_1525034 @ 5901-8393-7689

Amazon S3 > Buckets

Successfully created bucket "codepipelineartifactbucket123"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

Account snapshot - updated every 24 hours All AWS Regions [View Storage Lens dashboard](#)
Storage lens provides visibility into storage usage and activity trends. [Learn more](#)

General purpose buckets Directory buckets

General purpose buckets (1) Info All AWS Regions
Buckets are containers for data stored in S3.

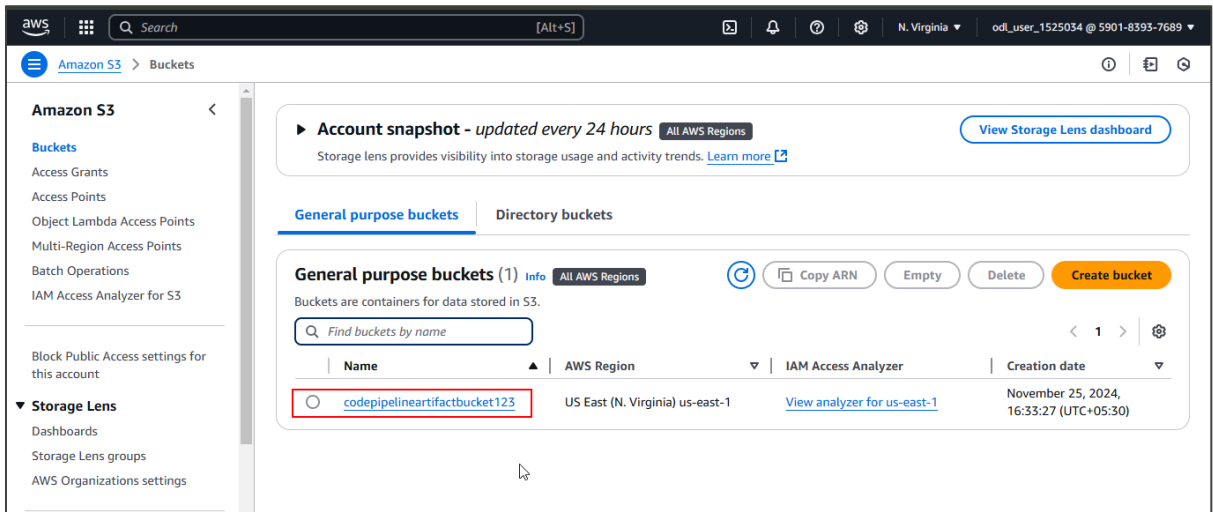
codepipeline 0 matches

Name AWS Region IAM Access Analyzer Creation date

No matches
We can't find a match.
[Clear filters](#)

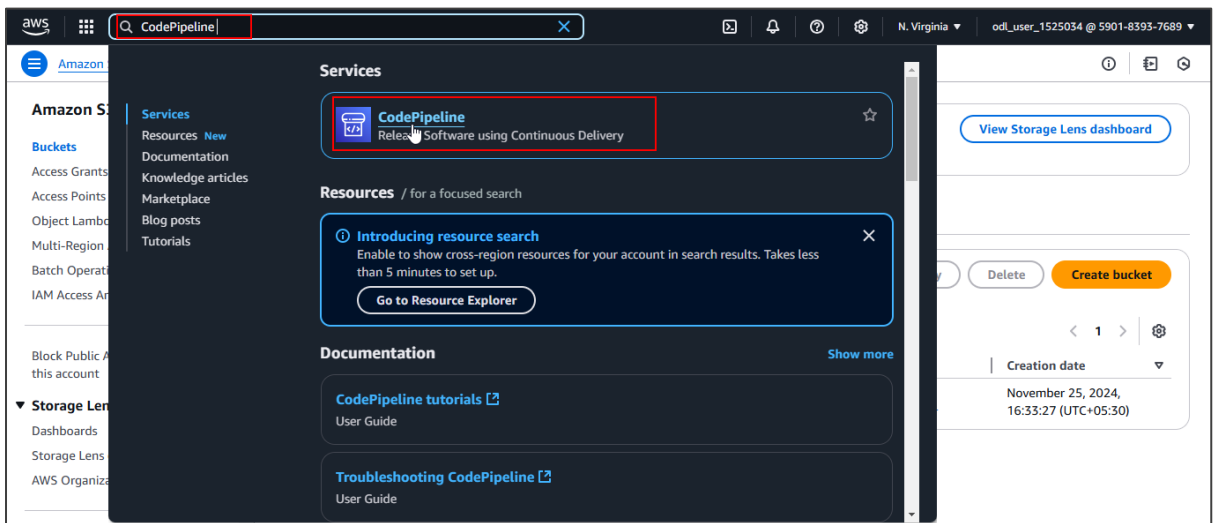
Copy ARN Empty Delete Create bucket

1.17 Navigate to the **Buckets** section of the **Amazon S3** service to verify the creation of the bucket

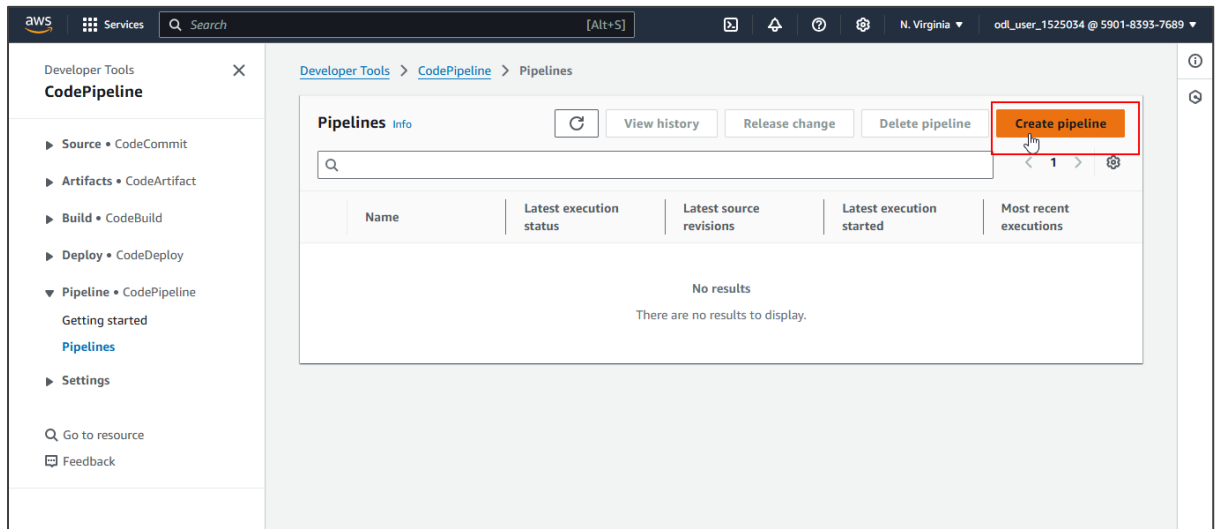


Step 2: Create a Continuous Integration pipeline

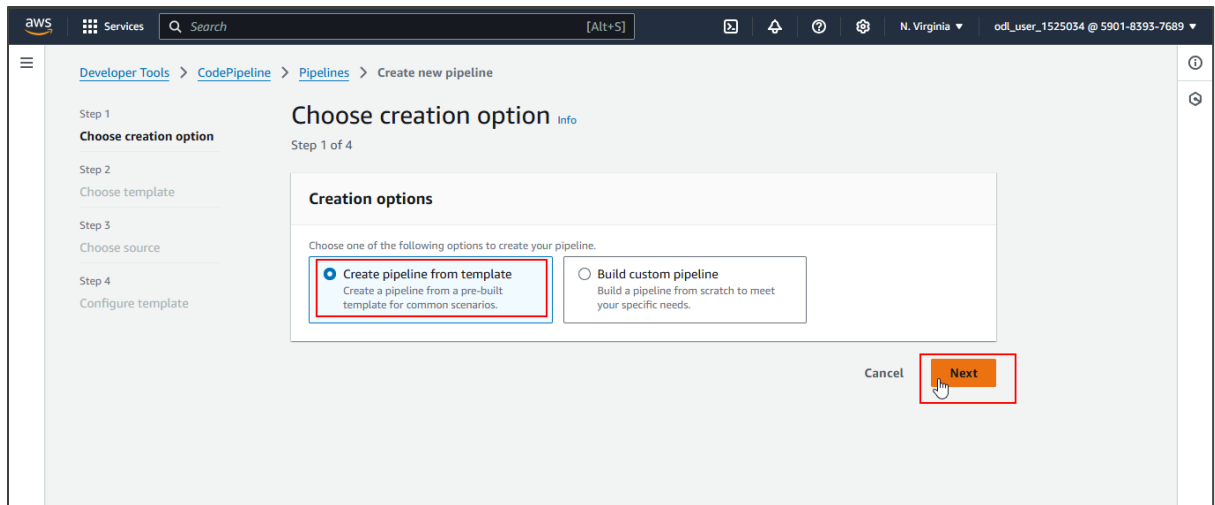
2.1 Search for and select the **CodePipeline** service



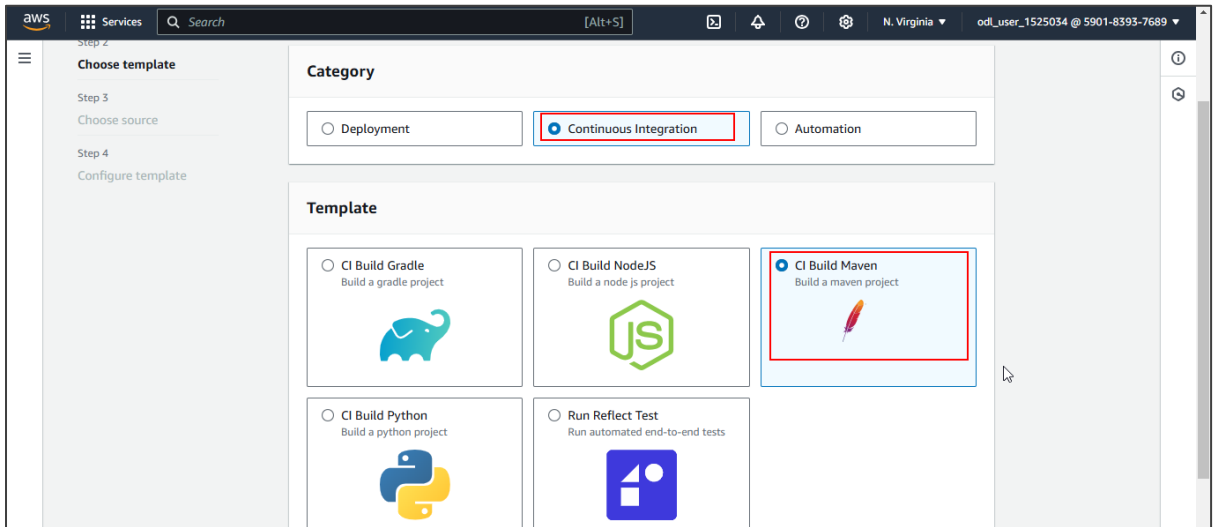
2.2 Click on the **Create pipeline**



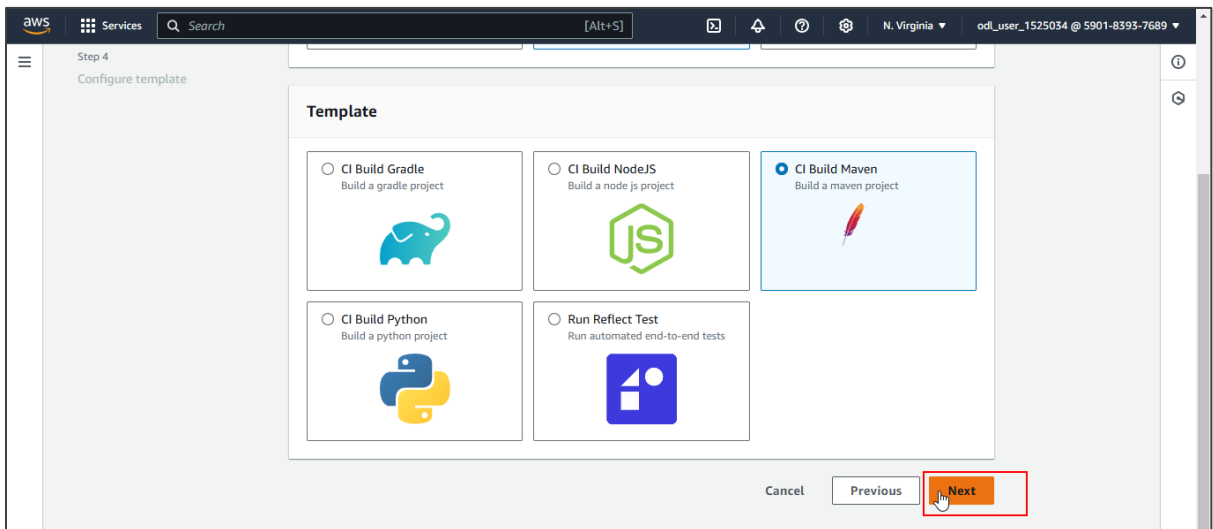
2.3 Select **Create pipeline from template** under the **Creation options** and click on **Next**



2.4 Select **Continuous Integration** under **Category** and **CI Build Maven** under the **Template** section



2.5 Scroll down and click on **Next**



2.6 Select **GitHub (via GitHub App)** under **Source provider** and click on **Connect to GitHub**

The screenshot shows the AWS CodePipeline console in the 'Choose source' step configuration. The breadcrumb navigation is 'Developer Tools > CodePipeline > Pipelines > Create new pipeline'. The left sidebar shows the steps: Step 1 (Choose creation option), Step 2 (Choose template), Step 3 (Choose source), and Step 4 (Configure template). The main content area is titled 'Choose source' and 'Step 3 of 4'. Under the 'Source' section, the 'Source provider' dropdown is set to 'GitHub (via GitHub App)'. Below this, the 'Connection' section has a search box and a 'Connect to GitHub' button. The 'Repository name' section has a search box. The 'Default branch' section has a text input field.

Developer Tools > CodePipeline > Pipelines > Create new pipeline

Step 1
Choose creation option

Step 2
Choose template

Step 3
Choose source

Step 4
Configure template

Choose source

Step 3 of 4

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

GitHub (via GitHub App)

Connection
Choose an existing connection that you have already configured, or create a new one and then return to this task.

or **Connect to GitHub**

Repository name
Choose a repository in your GitHub account.

Default branch
Default branch will be used only when pipeline execution starts from a different source or manually started.

2.7 Enter the **Connection name** as **connection** and click on **Connect to GitHub**

The screenshot shows the AWS CodePipeline console in the 'Create a connection' step configuration. The breadcrumb navigation is 'Developer Tools > Connections > Create connection'. The left sidebar shows the steps: Step 1 (Choose creation option), Step 2 (Choose template), Step 3 (Choose source), and Step 4 (Configure template). The main content area is titled 'Create a connection' and 'Step 3 of 4'. Under the 'Create GitHub App connection' section, the 'Connection name' input field is set to 'connection'. Below this, there is a 'Tags - optional' section. The 'Connect to GitHub' button is highlighted.

Create connection | CodePipeline | us-east-1 - Google Chrome

us-east-1.console.aws.amazon.com/codesuite/settings/connections/create?origin=codepipeline&providers=github®ion=us-east-1#

Developer Tools > Connections > Create connection

Create a connection

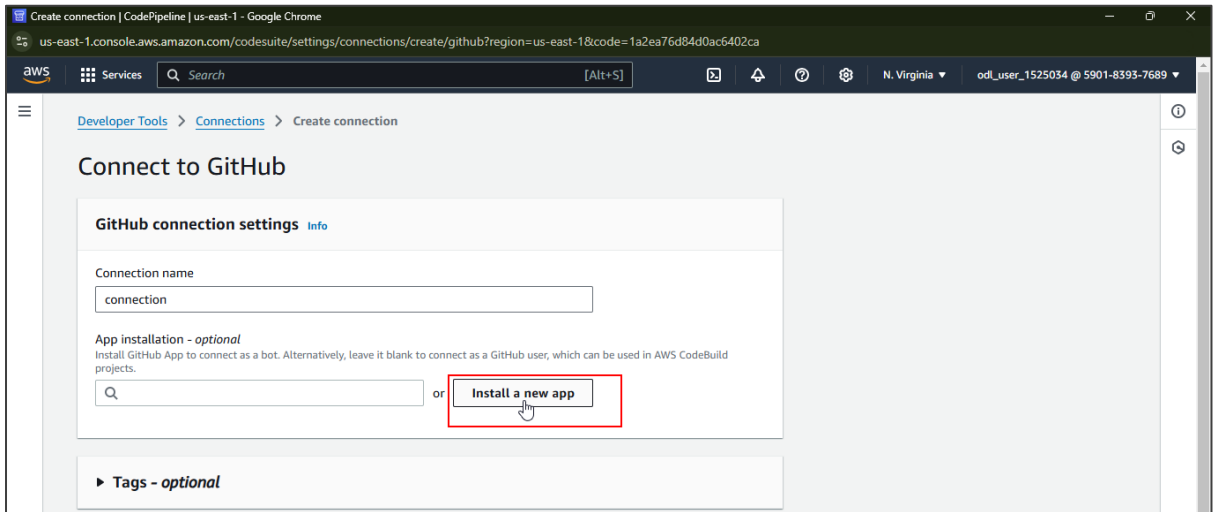
Create GitHub App connection

Connection name
connection

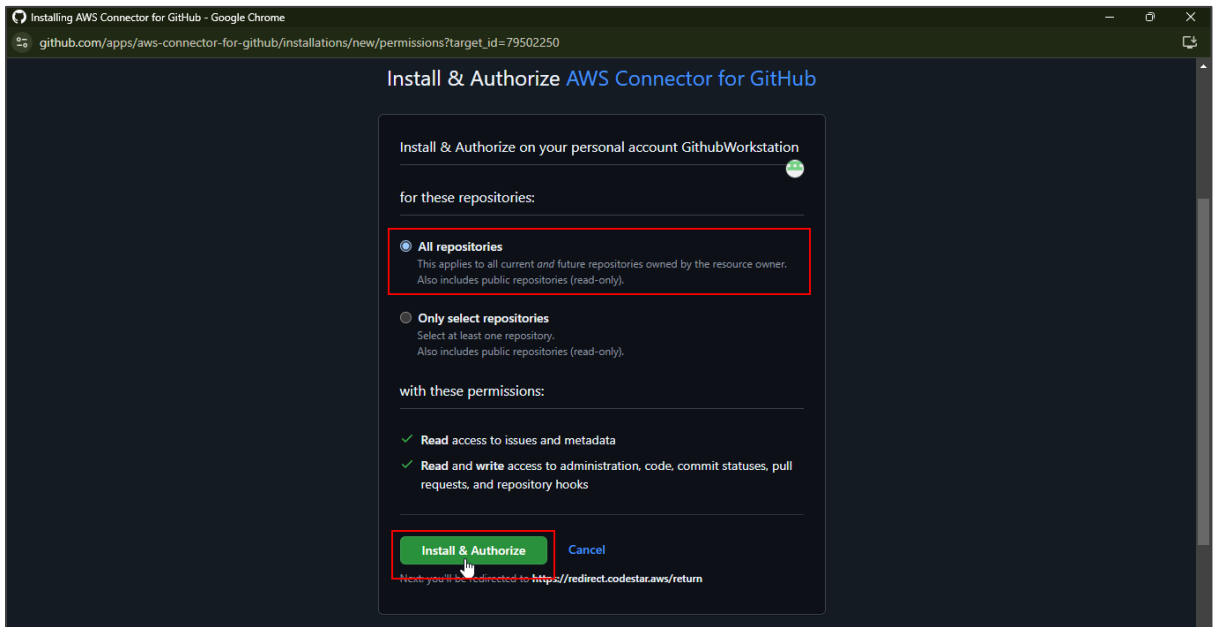
Tags - optional

Connect to GitHub

2.8 Click on **Install a new app**



2.9 Select **All repositories** and click on **Install & Authorize**



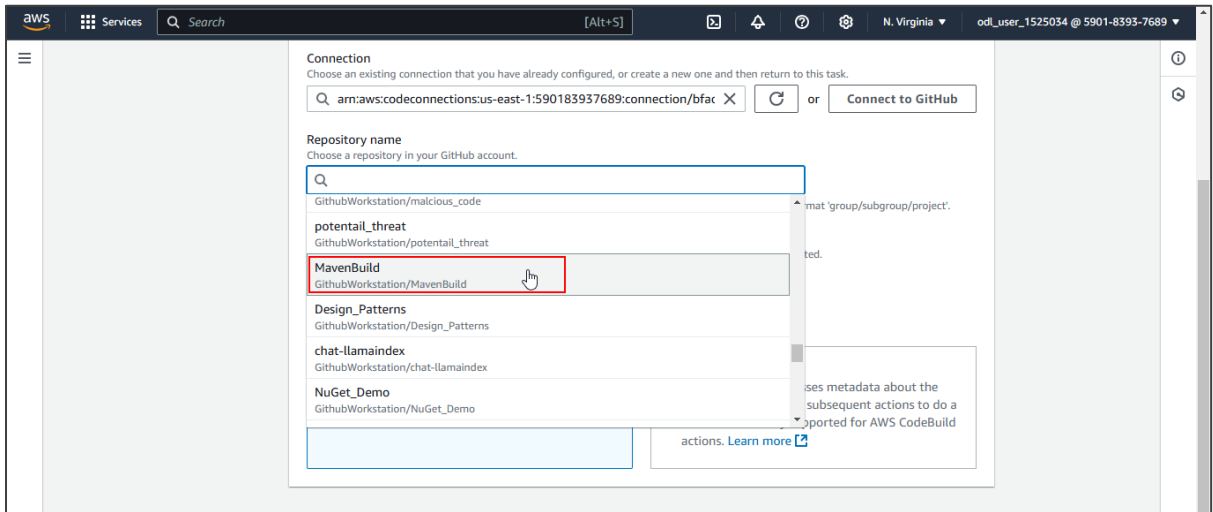
2.10 Select the available App installation ID and click on **Connect**

The screenshot shows the 'Create connection' page in the AWS CodePipeline console. The page title is 'Connect to GitHub'. Under 'GitHub connection settings', there is a 'Connection name' field with the value 'connection'. Below it, under 'App installation - optional', there is a text input field containing the App installation ID '57582014'. At the bottom right, there is an orange 'Connect' button. Red boxes highlight the App installation ID field and the 'Connect' button.

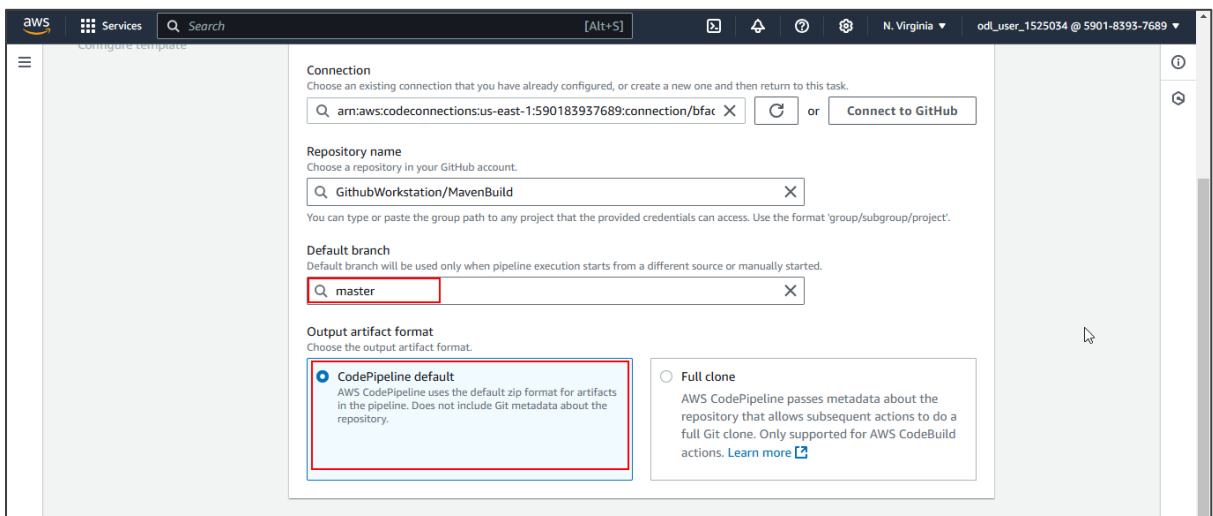
2.11 Navigate back to the Code Pipeline page and select the available connection in the **Connection** section

The screenshot shows the 'Choose source' page in the AWS CodePipeline console, Step 3 of 4. The 'Source provider' is set to 'GitHub (via GitHub App)'. In the 'Connection' section, the dropdown menu shows 'arn:aws:codeconnections:us-east-1:590183937689:connection/bfac'. To the right of the dropdown are a refresh icon and a 'Connect to GitHub' button. Below the connection field, there are input fields for 'Repository name' and 'Default branch'. The left sidebar shows the navigation steps: Step 1 (Choose creation option), Step 2 (Choose template), Step 3 (Choose source), and Step 4 (Configure template).

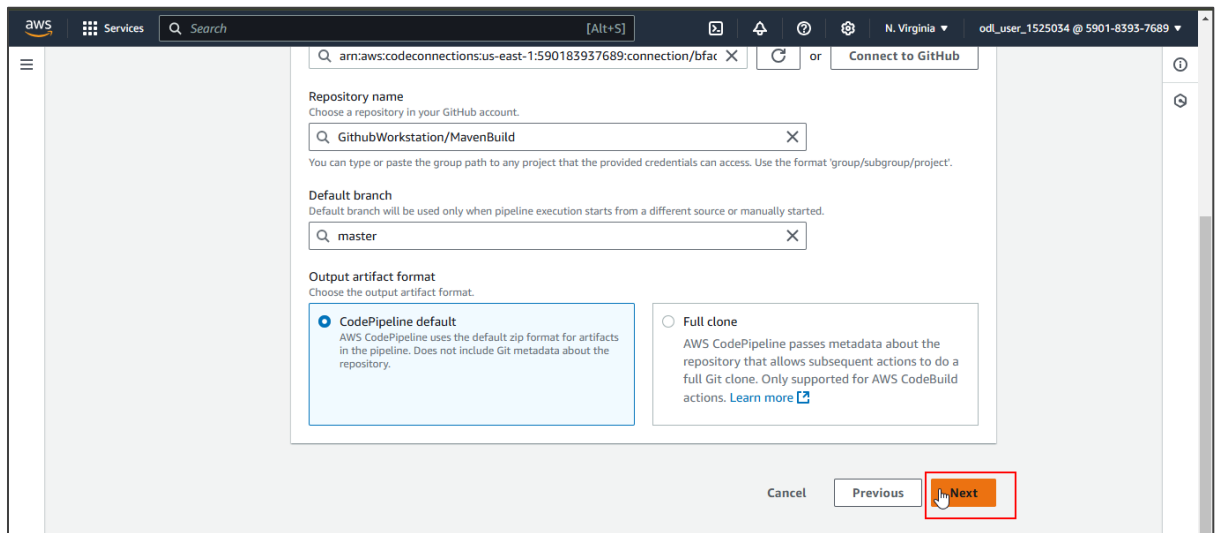
2.12 Select the **Repository name** as **MavenBuild** from the dropdown



2.13 Select the **Default branch** as **master** and **Output artifact format** as **CodePipeline default**



2.14 Scroll down and click on **Next**



The screenshot shows the AWS CodePipeline console interface. At the top, there's a search bar and navigation tabs. The main content area is titled 'Repository name' and 'Choose a repository in your GitHub account.' Below this, there's a text input field containing 'GithubWorkstation/MavenBuild'. A note below the input field states: 'You can type or paste the group path to any project that the provided credentials can access. Use the format 'group/subgroup/project'.' Below the input field, there's a 'Default branch' section with a text input field containing 'master'. At the bottom, there's an 'Output artifact format' section with two radio button options: 'CodePipeline default' (selected) and 'Full clone'. The 'CodePipeline default' option has a description: 'AWS CodePipeline uses the default zip format for artifacts in the pipeline. Does not include Git metadata about the repository.' The 'Full clone' option has a description: 'AWS CodePipeline passes metadata about the repository that allows subsequent actions to do a full Git clone. Only supported for AWS CodeBuild actions. [Learn more](#)'. At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Next'. The 'Next' button is highlighted with a red rectangle and a mouse cursor is clicking on it.

2.15 Enter the code given below in the **CI CodeBuildSpec** section

version: 0.2

phases:

build:

commands:

- echo "Starting maven build"
- mvn -B package --file pom.xml
- echo "Completed maven build successfully!"

artifacts:

files:

- target/*.war

discard-paths: 'yes'

encryption-key: 'arn:aws:kms:us-east-1:590184034797:key/fb356aee-f179-4426-a9ec-200649f3c997'

BranchName
The branch name to use with your CodeConnections connection.

CodePipelineName
The CodePipeline pipeline name that will build your maven source project.

CICodeBuildSpec
The CodeBuild project build spec maven configuration

```
1 version: 0.2
2 phases:
3   build:
4     commands:
5       - echo "Starting maven build"
6       - mvn -B package --file pom.xml
7       - echo "Completed maven build successfully!"
8   artifacts:
9     files:
10      - target/*.war
11      - discard-paths: 'yes'
12      - encryption-key: 'arn:aws:kms:us-east-1:590184034797:key/fb356aee-f179-4426-a9ec-200649f3c997'
13
```

RetentionPolicy
Define if you'd like the resource retained or deleted when the CloudFormation stack is deleted.

Delete

Note: Ensure you provide your specific encryption key ARN in the above code

2.16 Scroll down and click on **Create pipeline from template**

CICodeBuildSpec
The CodeBuild project build spec maven configuration

```
1 version: 0.2
2 phases:
3   build:
4     commands:
5       - echo "Starting maven build"
6       - mvn -B package --file pom.xml
7       - echo "Completed maven build successfully!"
8   artifacts:
9     files:
10      - target/*.war
11      - discard-paths: 'yes'
12      - encryption-key: 'arn:aws:kms:us-east-1:590184034797:key/fb356aee-f179-4426-a9ec-200649f3c997'
13
```

RetentionPolicy
Define if you'd like the resource retained or deleted when the CloudFormation stack is deleted.

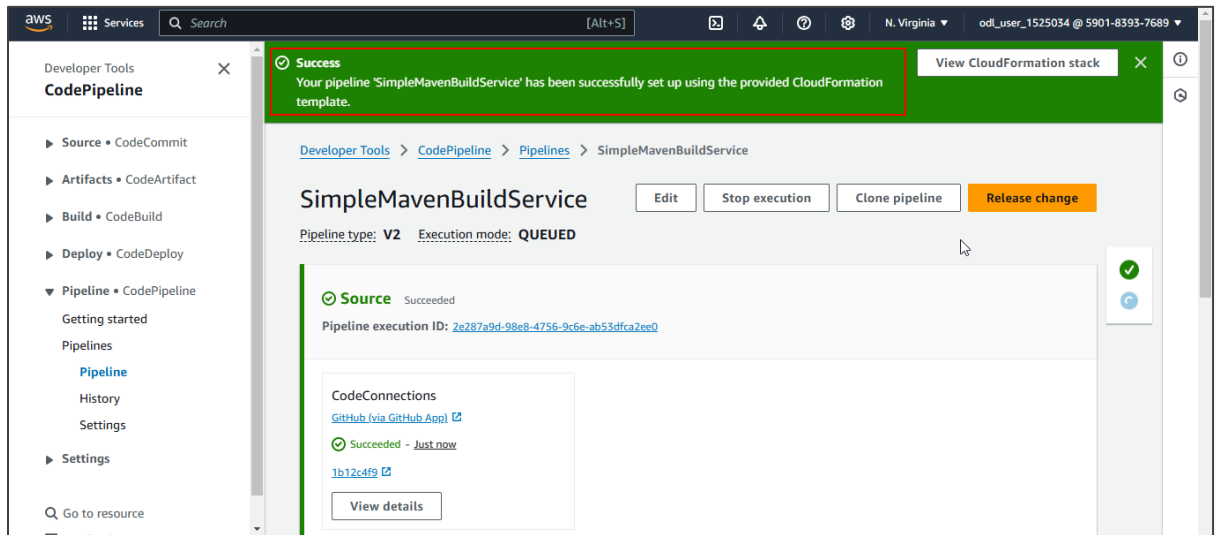
Delete

► Preview

Copy code

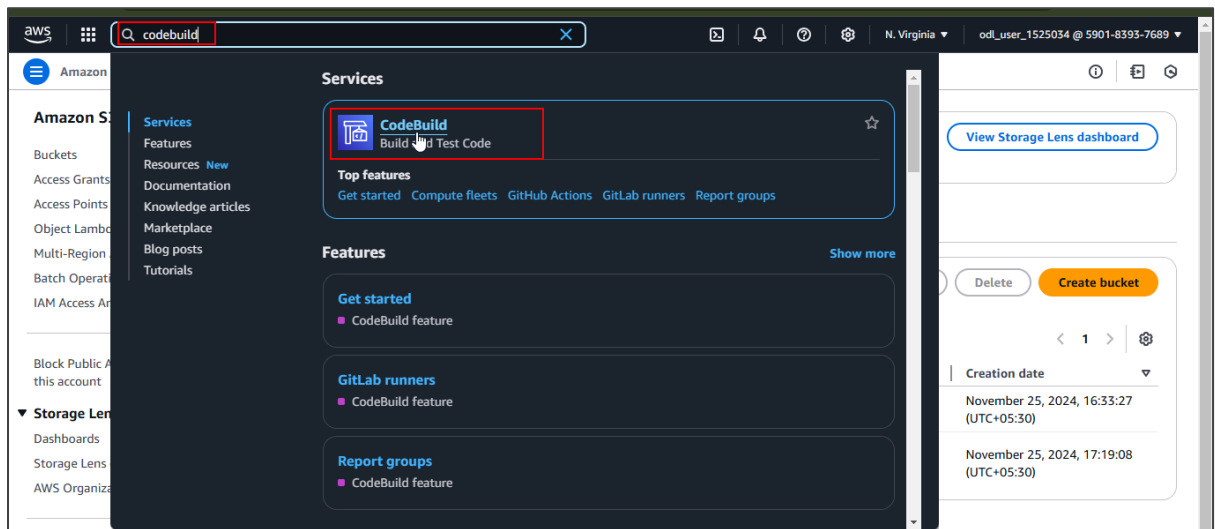
Cancel Previous **Create pipeline from template**

The pipeline is created successfully.

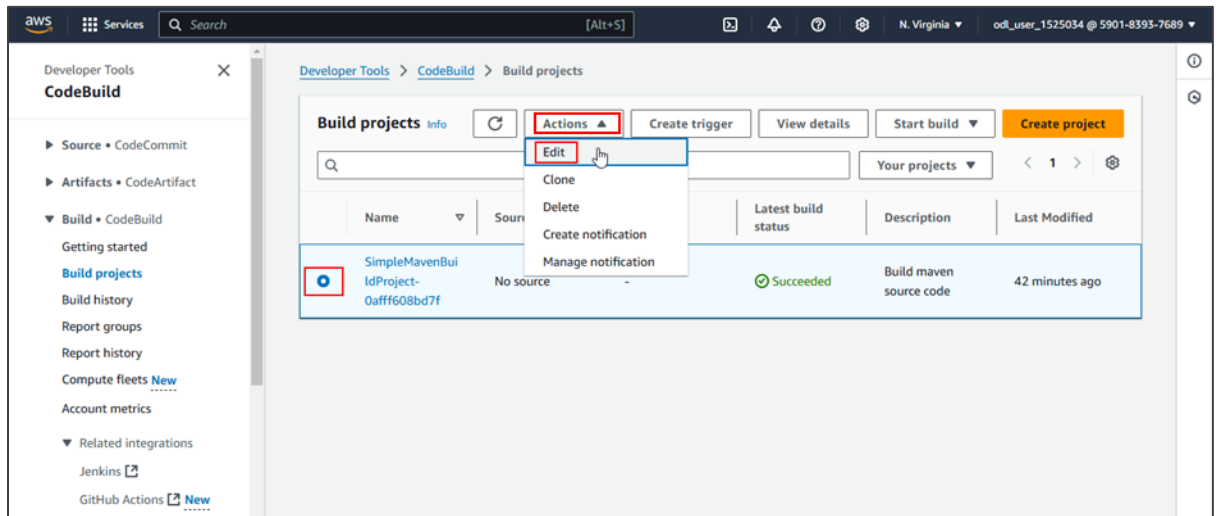


Step 3: Edit the pipeline and add the Artifacts section

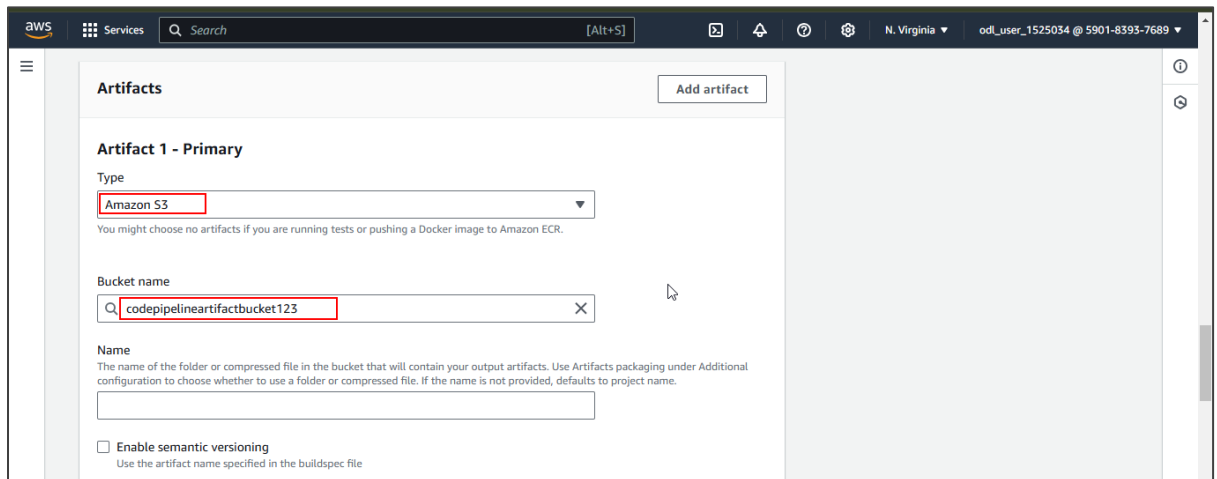
3.1 Search for and select the CodeBuild service



3.2 Select the Pipeline created in the previous step and click on **Edit** under the **Actions** tab



3.3 Scroll down to the **Artifacts** section and select the **Type** as **Amazon S3** and the Bucket name as **codepipelineartifactbucket123** which was created in the first step



3.4 Select the **Namespace type** as **None** and enter the **Encryption key** generated in step 1.14 and the **Cache type** as **No cache**

aws Services Search [Alt+S] N. Virginia odl_user_1525034 @ 5901-8393-7689

Namespace type - optional
None
Choose Build ID to insert the build ID into the path to the build output ZIP file or folder, e.g. MyPath/MyBuildID/MyArtifact.zip. Otherwise, choose None.

Artifacts packaging
☒ None
The artifact files will be uploaded to the bucket.
☐ Zip
AWS CodeBuild will upload artifacts into a compressed file that is put into the specified bucket.

☐ Disable artifact encryption
Disable encryption if using the artifact to publish a static website or sharing content with others

▼ Additional configuration
Cache, encryption key

Encryption key - optional
Provide the AWS KMS customer master key used to encrypt this build's output artifacts. The default is your AWS-managed customer master key for S3.
arn:aws:kms:us-east-1:590183937689:alias/aws/s3
arn:aws:kms:<region-ID>:<account-ID>:key/<key-ID>

Cache type
No cache

3.5 Scroll down and click on **Update project**

aws Services Search [Alt+S] N. Virginia odl_user_1525034 @ 5901-8393-7689

Stream name prefix - optional
The prefix of the stream name of the CloudWatch Logs.

S3
☐ S3 logs - optional
Checking this option will upload build output logs to S3.

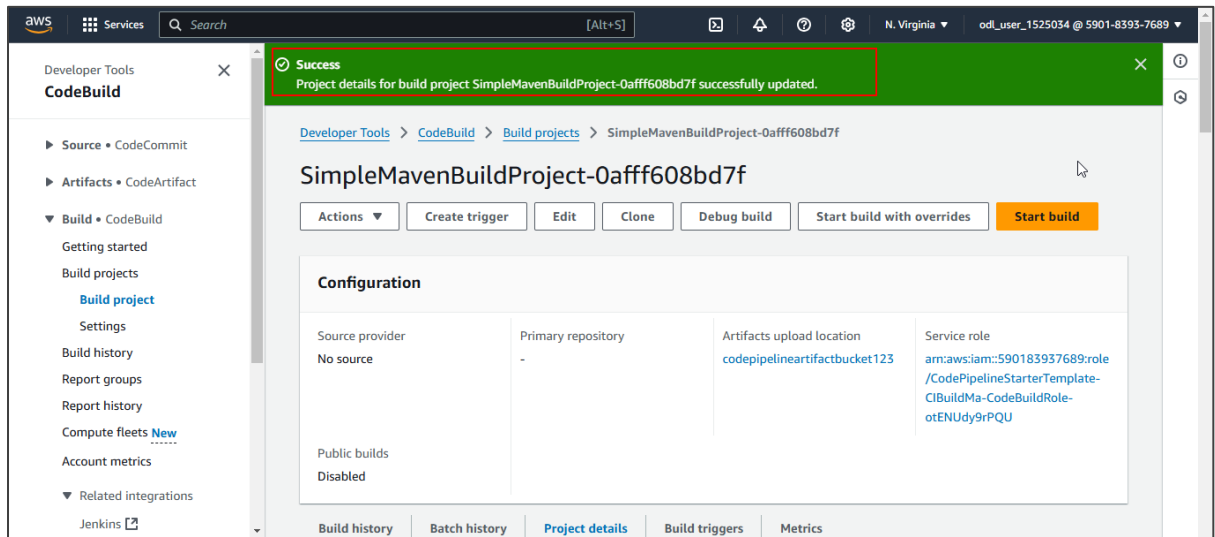
Service role permissions

Service role
Choose an existing service role from your account
arn:aws:iam::590183937689:role/CodePipelineStarterTemplate-CIBuildMa-Coc X

☒ Allow AWS CodeBuild to modify this service role so it can be used with this build project

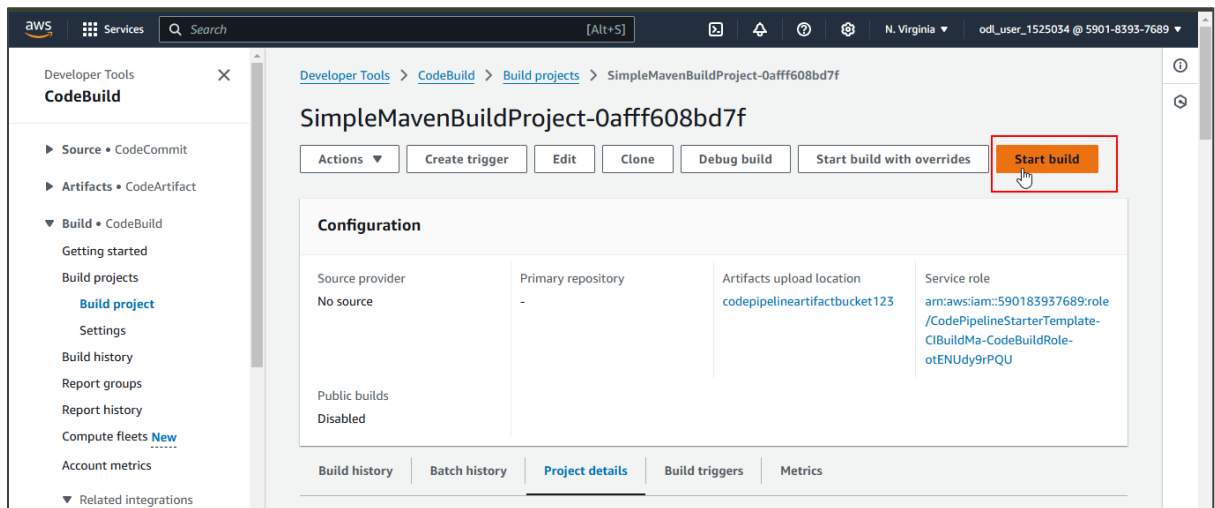
Cancel **Update project**

The pipeline is updated successfully.

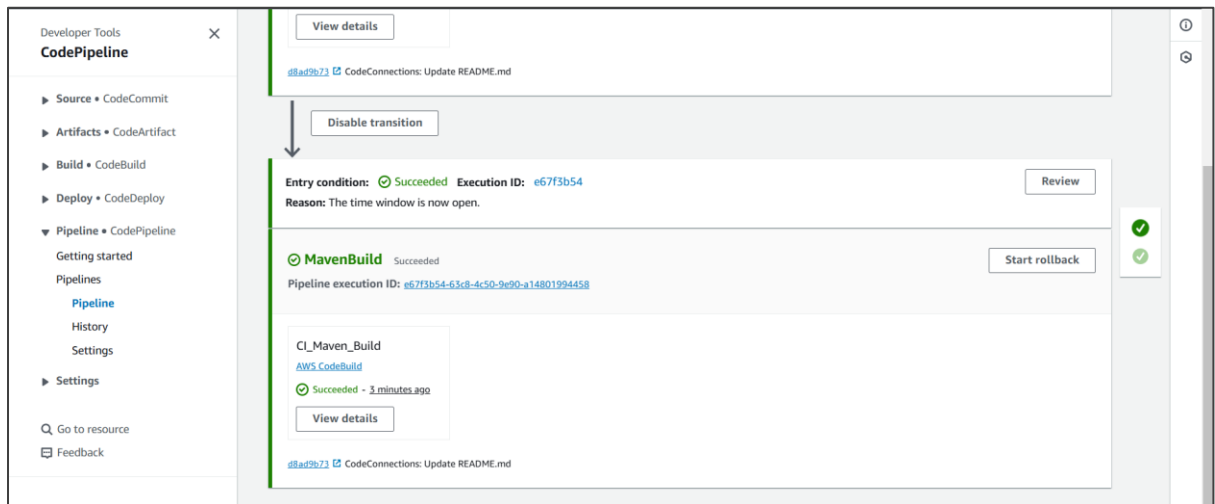


Step 4: Test the updated pipeline

4.1 Navigate back to the pipeline in the **Build projects** and click on **Start build**



The pipeline starts building, and the changes are successfully implemented in the new build.



By following the above steps, you have successfully created a secure Continuous Integration pipeline using AWS CodePipeline, configured artifact encryption with AWS KMS, and built a Maven project while ensuring the secure storage and management of outputs in Amazon S3.